Notes: Taylor (JF, 2010)

Why Are CEOs Rarely Fired? Evidence from Structural Estimation

Contents

1 Big picture

- **Empirical puzzle:** CEO dismissals are rare (roughly a few percent per year), yet CEO performance differs and boards appear to learn about managers over time.
- **Economic interpretation:** "Rare firing" can arise from (i) *learning* (boards are uncertain about ability and update beliefs) and (ii) *turnover costs* (real costs to the firm plus board/private costs that create CEO entrenchment).
- **Methodological move:** estimate a *fully dynamic* CEO retention problem with a *persistent* performance signal, using *structural* parameters disciplined by moments in the data (Simulated Method of Moments, SMM).
- **Outputs:** (i) the implied magnitudes of firm turnover costs and board "personal" turnover costs (entrenchment), (ii) the implied learning speed and signal precision, and (iii) counterfactual welfare effects on shareholder value.

Throughout these notes, we keep notation consistent with the paper (Table I in the paper).

2 Data objects and empirical measurement (what is observed)

2.1 CEO spells, forced vs unforced

A *CEO spell* is the sequence of years a given CEO remains in office. A spell ends either by:

- **Forced** turnover (dismissal / firing), or
- **Unforced** turnover (voluntary exit, retirement, etc.).

The empirical classification uses a standard algorithm (discussed in the paper) and is varied in robustness checks.

2.2 Profitability and industry adjustment

Let Y_{it} denote firm i's profitability rate in year t (in % of assets per year). Let v_{it} denote an industry profitability benchmark (industry average profitability). Define *excess (firm-specific) profitability*:

$$y^*_{it} \equiv Y_{it} - v_{it}. \tag{1}$$

Intuitively, y^*_{it} is what the CEO can (in principle) influence after removing broad industry conditions.

2.3 Key empirical facts (Table II)

Table II (paper) summarizes the sample and the basic turnover rates:

- Full sample: 7,325 firm-years and 981 CEO spells.

- Forced successions: 168 out of 981 (17.1% of successions), which corresponds to 2.29% forced turnover per firm-year.
- Forced turnover rises over time (e.g., from 7.8% of successions in 1970–1974 to 24.7% in 2005–2006).
- Profitability is volatile: excess profitability has sizable dispersion across firm-years; CEO spell lengths are long on average, but shorter for spells that end in dismissal.

3 The structural model

3.1 Timeline and primitives

Time is discrete ($t = 0, 1, 2, \dots$) and one period is a year in the empirical implementation.

There is a firm with book assets B_t at the beginning of period t. For tractability, the paper assumes profits are paid out as dividends so that **assets are constant over time**:

$$B_t \equiv B \quad \text{for all } t.$$

A CEO can be replaced at the beginning of each period. Also, after having served τ periods, a CEO may exit voluntarily with an *exogenous* probability $f(\tau)$ (estimated from the data and treated as an input).

3.2 Profitability process and CEO ability

Profitability has an industry component plus a firm-specific component. Let v_t denote industry profitability and y_t denote firm-specific profitability *net of the firm's turnover cost.* Turnover (forced or voluntary) occurs in period t if the indicator $\mathbf{1}\{\text{turn}_t\} = 1$.

Observed profitability (Equation (1) in the paper).

$$Y_t = v_t + y_t - \mathbf{1}\{\text{turn}_t\}\, c^{(\text{firm})}. \tag{1}$$

Here $c^{(\text{firm})}$ is the *real* turnover cost borne by shareholders (as a fraction of assets).

Firm-specific profitability dynamics (Equation (2) in the paper). The key state variable y_t mean reverts toward the CEO's latent ability α:

$$y_t = y_{t-1} + \phi(\alpha - y_{t-1}) + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2). \tag{2}$$

Interpretation:

- α is constant over the CEO's tenure.

- $\phi \in (0, 1)$ controls persistence: smaller ϕ means *more persistence* (slow mean reversion), so past performance continues to influence y_t.
- ε_t captures idiosyncratic profitability noise.

3.3 Turnover costs and the meaning of "entrenchment"

There are **two** turnover costs:

1. **Firm cost $c^{(\text{firm})}$:** affects profits directly (severance, search fees, disruption, etc.).
2. **Board personal cost $c^{(\text{pers})}$:** does *not* affect profits, but enters the board's objective (stress, reputational consequences, loss of CEO as ally, etc.).

Baseline assumption: forced and voluntary turnover are equally costly (a robustness check sets voluntary turnover costs to zero).

3.4 Board objective, discounting, and the effective personal cost

The board chooses a firing decision each period to maximize discounted utility.

Dynamic objective (Equation (3)). Let $d_t \in \{\text{fire}, \text{keep}\}$ denote the board action. The board chooses a policy $\{d_{t+s}\}_{s\geq 0}$ to maximize

$$U_t = \mathbb{E}_t\left[\sum_{s=0}^{\infty} \beta^s u_{t+s}\right], \qquad 0 < \beta < 1. \tag{3}$$

Per-period utility (Equation (4)). Since profits in dollars equal $B\,Y_t$, per-period utility is:

$$u_t = \kappa\, B\, Y_t \;-\; \mathbf{1}\{\text{turn}_t\}\, B\, c^{(\text{pers})}. \tag{4}$$

Here $\kappa > 0$ is "utils per dollar" of firm profits and scales how much the board internalizes shareholder value.

Decomposition (Equation (5)). Substituting (4) into (3),

$$U_t = \kappa\, \mathbb{E}_t\left[\sum_{s=0}^{\infty} \beta^s\, B\, Y_{t+s}\right] - \mathbb{E}_t\left[\sum_{s=0}^{\infty} \beta^s\, \mathbf{1}\{\text{turn}_{t+s}\}\, B\, c^{(\text{pers})}\right]. \tag{5}$$

This highlights the central conceptual object:

$$\underbrace{\frac{c^{(\text{pers})}}{\kappa}}_{\text{"effective personal cost" in \% of assets}}$$

Only the *ratio* $c^{(\mathrm{pers})}/\kappa$ is identified (utility is defined up to scale), so the paper estimates **effective personal cost** directly.

3.5 Information and Bayesian learning about CEO ability

3.5.1 Prior

When a new CEO is hired, his ability is drawn from the talent pool:

$$\alpha \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

Here μ_0 is the prior mean ability; σ_0 is both (i) the board's initial uncertainty about a new CEO and (ii) the cross-CEO dispersion of ability in the hiring pool.

3.5.2 Signals

Each period, the board observes two signals:

1. **Firm-specific profitability** y_t (which depends on α and noise ε_t).
2. **Additional signal** z_t capturing other public/private information:

$$z_t \sim \mathcal{N}(\alpha, \sigma_z^2),$$

   independent of profitability innovations.

3.5.3 A key transformation: turning y_t into a direct noisy signal of ability

From (2),

$$y_t - (1-\phi)y_{t-1} = \phi\alpha + \varepsilon_t \quad \Rightarrow \quad x_t \equiv \frac{y_t - (1-\phi)y_{t-1}}{\phi} = \alpha + \frac{\varepsilon_t}{\phi}.$$

Thus x_t is an unbiased signal of α with noise variance $\sigma_\varepsilon^2/\phi^2$.

The paper uses this transformation explicitly for identification; a special case is written as Equation (6).

3.5.4 Posterior updating formulas (derivation)

Let the prior at the start of period t be $\alpha \mid \mathcal{I}_{t-1} \sim \mathcal{N}(m_{t-1}, s_{t-1}^2)$. Given independent Gaussian signals

$$x_t = \alpha + \nu_t, \quad \nu_t \sim \mathcal{N}(0, \sigma_\varepsilon^2/\phi^2), \qquad z_t = \alpha + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_z^2),$$

conjugacy implies the posterior is Gaussian with precision additivity:

$$\frac{1}{s_t^2} = \frac{1}{s_{t-1}^2} + \frac{\phi^2}{\sigma_\varepsilon^2} + \frac{1}{\sigma_z^2}. \tag{2}$$

and posterior mean

$$m_t = s_t^2 \left(\frac{m_{t-1}}{s_{t-1}^2} + \frac{\phi^2}{\sigma_\varepsilon^2} x_t + \frac{1}{\sigma_z^2} z_t \right). \tag{3}$$

Proof sketch. Because (α, x_t, z_t) are jointly normal and the signal noises are independent, the posterior is normal and can be obtained by completing the square in the log posterior density:

$$\log p(\alpha \mid x_t, z_t) = \text{const} - \frac{(\alpha - m_{t-1})^2}{2s_{t-1}^2} - \frac{(x_t - \alpha)^2}{2(\sigma_\varepsilon^2/\phi^2)} - \frac{(z_t - \alpha)^2}{2\sigma_z^2}.$$

Collect quadratic terms in α to read off the posterior precision and mean, yielding (2)–(3). □

3.6 Dynamic programming representation and the firing threshold

3.6.1 State variables

At the beginning of period t, the economically relevant state can be represented by:

$$\text{state}_t = (\tau_t,\ y_{t-1},\ m_{t-1},\ s_{t-1}^2),$$

where τ_t is CEO tenure (years already served), and (m_{t-1}, s_{t-1}^2) summarize beliefs about α.

3.6.2 Bellman equation (conceptual form)

Let $V(\tau, y, m, s^2)$ be the value for the board at the start of a period. Then

$$V(\tau, y, m, s^2) = \max \left\{ V^{\text{keep}}(\tau, y, m, s^2),\ V^{\text{fire}}(\tau, y, m, s^2) \right\},$$

where:

- V^{keep} includes current expected utility plus continuation value under updated beliefs and next-period state evolution.
- V^{fire} subtracts turnover costs (firm and personal), resets beliefs to the prior (μ_0, σ_0^2) for the new CEO, and then continues.

The paper solves this numerically.

3.6.3 Threshold policy (why "fire when belief is low")

The paper states (and uses) the property that the board fires when the posterior mean drops below a cutoff that depends on tenure and parameters.

Proposition 1 (Monotone cut-off in posterior mean (economic argument))**.** *Fix (τ, y, s^2). Suppose the value of keeping the CEO is increasing in the posterior mean m (higher expected ability raises* *expected future profitability), and the value of firing depends on m only through the current state but resets to the prior. Then there exists a cutoff $\bar{m}(\tau, y, s^2)$ such that:*

$$\textit{keep} \iff m \geq \bar{m}(\tau, y, s^2), \qquad \textit{fire} \iff m < \bar{m}(\tau, y, s^2).$$

Proof sketch. Define the value difference $\Delta(m) \equiv V^{\text{keep}}(\tau, y, m, s^2) - V^{\text{fire}}(\tau, y, m, s^2)$. Under the stated monotonicity, $\Delta(m)$ is increasing in m. Thus the set $\{m : \Delta(m) \geq 0\}$ is an interval $[\bar{m}, \infty)$ for some cutoff $\bar{m}$, implying a threshold rule. □

3.6.4 Comparative statics (linking parameters to firing behavior)

The model delivers robust directional predictions (illustrated in Figure 1–3 in the paper):

- Higher **total turnover cost**
$$c \equiv c^{(\text{firm})} + \frac{c^{(\text{pers})}}{\kappa}$$
lowers the firing threshold (board tolerates lower perceived ability), reducing firing rates.

- Learning is slower (and firing occurs later) when information is less precise: higher σ_z (noisier extra signal), higher σ_ε (noisier profits), or smaller ϕ (more persistence, so current profits reveal ability more slowly).

- Larger σ_0 (greater dispersion/uncertainty about new CEOs) makes early-tenure firing more likely because boards expect larger potential gains from replacing a bad CEO.

4 Estimation: parameters, moments, and identification

4.1 Parameter vector

The paper estimates seven parameters:

$$\theta = \left(c^{(\text{firm})},\ c^{(\text{pers})}/\kappa,\ \mu_0,\ \sigma_0,\ \phi,\ \sigma_\varepsilon,\ \sigma_z\right),$$

while fixing β (baseline $\beta = 0.9$) and using the empirical voluntary-turnover hazard $f(\tau)$ as an input.

4.2 Identification logic (how moments map to parameters)

The key idea is to select moments that separately discipline:

- the **time-series dynamics** of profitability (pins down ϕ and σ_ε),
- the **event-time pattern** of profitability around dismissals (pins down σ_z and $c^{(\text{firm})}$),
- the **frequency/timing of firings** by tenure (pins down the total turnover cost c),
- the **cross-CEO dispersion** in profitability (separates σ_0 from σ_ε).

4.3 Pooled regression moments (Equation (8))

The first set of moments comes from a pooled regression that describes profitability dynamics and event-time profitability around forced turnover:

$$y^*_{it} = \lambda_0 + \lambda_1 y^*_{i,t-1} + \sum_{k=-2}^{2} \gamma_k \, D^{(k)}_{it} + \delta_{it}, \tag{8}$$

where $D^{(k)}_{it}$ is an indicator for whether firm i's CEO is *forced out* at the end of year $t+k$. Interpretation of coefficients:

- λ_0 helps identify μ_0 (average profitability under a new draw from the CEO pool).
- λ_1 helps identify persistence ϕ (mapping from the structural persistence to an empirical AR(1)).
- $\gamma_{-2}, \gamma_{-1}, \gamma_0, \gamma_1, \gamma_2$ describe the *shape* of profitability in event time around dismissals, disciplining learning speed and the effect of turnover costs.
- $\mathrm{Var}(\delta)$ helps identify profitability noise.

4.4 Persistence-adjusted profitability and Equation (6)

The paper uses a transformation that removes predictable persistence, making the remaining variation closer to "ability + noise."

From the structural law of motion (2),

$$x_t \equiv \frac{y_t - (1-\phi)y_{t-1}}{\phi} = \alpha + \frac{\varepsilon_t}{\phi}.$$

A sample version uses the AR(1) estimate $\hat{\lambda}_1$ from (8):

$$\widehat{X}_{it} = \frac{y^*_{it} - \hat{\lambda}_1 y^*_{i,t-1}}{1 - \hat{\lambda}_1}.$$

Equation (6) in the paper illustrates the key variance decomposition in a special case:

$$X_{i,0} = \frac{y_{i,0} - y_{i,-1}}{\phi} + y_{i,-1} = \alpha_i + \frac{\varepsilon_{i,0}}{\phi}. \tag{6}$$

Therefore,

$$\mathrm{Var}(X_{i,0}) = \sigma_0^2 + \frac{\sigma_\varepsilon^2}{\phi^2}.$$

Once ϕ and σ_ε are disciplined by time-series moments, the cross-CEO dispersion moment identifies σ_0.

4.5 Turnover hazard-rate moments

The model generates predicted forced-turnover hazard rates at different tenure groups $h(1–2)$, $h(3–4)$, $h(5–7)$, and $h(8+)$. These moments are strongly informative about the total turnover cost c because increasing c shifts the firing threshold downward at all tenures, reducing all hazard rates.

4.6 SMM objective function (Equation (7))

Let $\widehat{M}$ be the vector of empirical moments and let $\widehat{m}_s(\theta)$ be the corresponding moment vector computed from the s-th simulated sample under parameter θ. The SMM estimator solves:

$$\widehat{\theta} \in \arg\min_{\theta} \left(\widehat{M} - \frac{1}{S} \sum_{s=1}^{S} \widehat{m}_s(\theta) \right)' W \left(\widehat{M} - \frac{1}{S} \sum_{s=1}^{S} \widehat{m}_s(\theta) \right), \tag{7}$$

where W is the efficient weighting matrix (inverse of the estimated covariance of $\widehat{M}$). The paper uses $S = 20$ simulated samples (each with 981 CEO spells, matching the empirical sample size) and uses simulated annealing to avoid local minima.

5 Results

5.1 Figure 1: hazard-rate comparative statics (model prediction)

Figure 1 (paper) shows predicted forced-dismissal hazard rates by tenure under different parameter values:

- **Top panel: varying total turnover cost c.** When c is low, early-tenure hazard is high because it is cheap to replace poorly performing CEOs. As c rises, hazard rates fall and shift later in tenure: boards require stronger evidence of low ability before paying the cost of turnover.
- **Bottom panel: varying σ_0.** Higher σ_0 (more dispersion/uncertainty about ability in the CEO pool) increases the value of replacing a bad CEO, raising early-tenure hazards. If all CEOs are similar (low σ_0), firing is rarely worth it.

5.2 Figure 2: profitability and beliefs around a dismissal (mechanism)

Figure 2 (paper) plots, in simulated data:

- average excess profitability around event time 0 (the firing year), and
- average posterior mean beliefs about the CEO's ability.

The key mechanism is visible:

1. Profitability declines as negative signals arrive and beliefs fall.

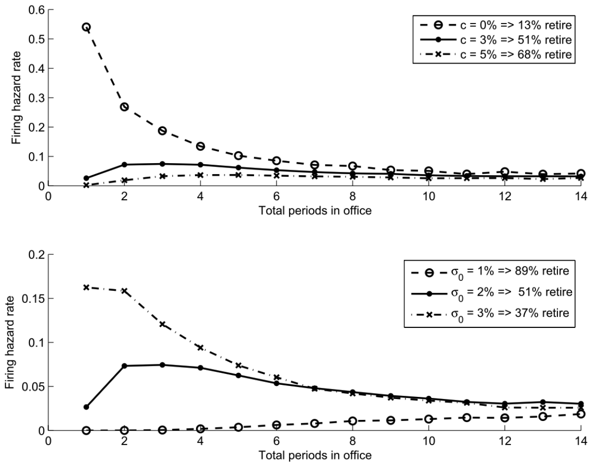

Figure 1. Predicted CEO dismissal hazard rates. This figure shows the hazard rates for CEO dismissals at different tenure levels τ. The hazard rate for tenure τ is the probability that the CEO will be fired after his τth period in office, conditional on him surviving to period τ. The legends indicate the percentage of CEOs who reach retirement after 15 periods without being fired. The top panel shows hazard rates for three different values of the total turnover cost, $c = c^{(firm)} + c^{(pers)}/\kappa$. The bottom panel shows hazard rates for three values of σ_0, the standard deviation of CEO ability in the pool of replacement CEOs. These results are from simulations of the model using parameter values $\beta = 0.9$, $\mu_0 = 1\%$, $\sigma_0 = 2\%$, $\sigma_\epsilon = 3\%$, $c = 3\%$, $\phi = 0.12$, and $\sigma_z = 7\%$; voluntary turnover occurs after (and only after) completing 15 periods in office.

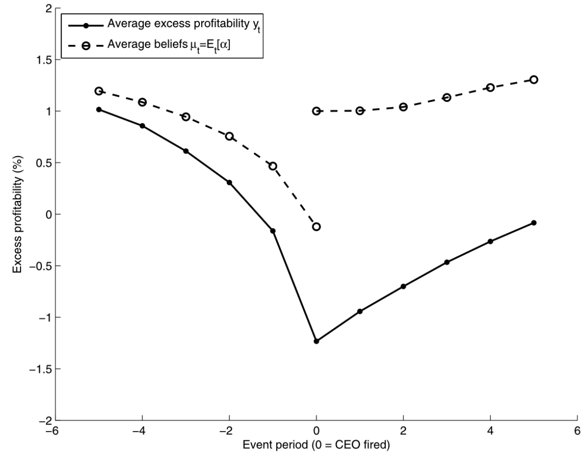

Figure 2. Predicted profitability around CEO dismissals. This figure plots average excess profitability in event time around CEO dismissals. Variable definitions are in Table I. The dashed line shows the average posterior mean beliefs about the CEO's ability. Results are from 100,000 consecutive CEO spells in a simulated firm. Simulations use the following parameter values: $\beta = 0.9$, $\mu_0 = 1\%$, $\sigma_0 = 2\%$, $\sigma_\epsilon = 3\%$, $c = 3\%$, $\phi = 0.12$, and $\sigma_z = 7\%$; voluntary turnover occurs after (and only after) completing 15 periods in office. Also, there are is no firm turnover cost, so $c^{(firm)} = 0$ and hence $c = c^{(pers)}/\kappa$.

2. Firing occurs when beliefs reach the threshold.

3. After replacement, beliefs jump upward because the new CEO is drawn from the prior distribution, and profitability begins recovering slowly (especially when ϕ is small, i.e., persistence is high).

5.3 Figure 3: which parameters change the event-time pattern

Figure 3 (paper) shows comparative statics for event-time profitability:

- **Top panel: σ_z (precision of additional information).** When σ_z is low (precise extra signal), boards learn faster and remove bad CEOs earlier, changing the depth and slope of profitability declines before firing. When σ_z is high (noisy extra signal), beliefs move more slowly and firings occur later, typically after a more extended deterioration in observed profitability.

- **Bottom panel: splitting total turnover cost into firm cost vs personal cost.** Holding total cost c fixed but raising $c^{(\text{firm})}$ creates a sharper mechanical profitability drop at the turnover date (because $c^{(\text{firm})}$ directly subtracts from Y_t in (1)).

5.4 Table I: definitions (how to keep notation straight)

Table I is a *notation dictionary*. For reading the rest:

- Y_t is overall profitability; v_t is industry profitability; $y_t^* = Y_t - v_t$ is excess profitability.

- y_t is firm-specific profitability net of firm turnover costs.

- α is CEO ability; (μ_0, σ_0) are prior mean and dispersion.

Table I

Variable and Subsample Definitions

This table contains definitions of the variables, parameters, empirical measures, and subsamples used most frequently throughout the paper. The text contains more detailed definitions.

Notation	Definition (units)
	Panel A: Model Parameters and Variables
Y_t	Firm profitability in year t (% of assets p.a.)
v_t	Industry average profitability in year t (% of assets p.a.)
y_t^*	Firm-specific profitability = excess profitability = $Y_t - v_t$ (% of assets p.a.)
y_t	Firm-specific profitability net of firm turnover cost (% of assets p.a.)
X_{it}	Persistence-adjusted firm-specific profitability (% of assets p.a.)
α	CEO ability (% of assets p.a.)
$c^{(firm)}$	CEO turnover cost to the firm (% of assets)
$c^{(pers)}$	Personal CEO turnover cost to the board (utils per \$, % of assets)
c	Total turnover cost = $c^{(firm)} + c^{(pers)}/\kappa$ (% of assets)
κ	Board utils per dollar of firm profits (utils per \$)
$c^{(pers)}/\kappa$	Effective personal turnover cost (% of assets)
μ_0	Mean of board's prior beliefs about new CEO's ability (% of assets p.a.)
σ_0	Standard deviation of prior beliefs about new CEO's ability (% of assets p.a.)
ϕ	Controls persistence in firm-specific profitability
σ_ϵ	Time-series volatility of firm-specific profitability (% of assets p.a.)
σ_z	Standard deviation of additional signal about CEO ability (% of assets p.a.)
β	Board's and investors' discount factor
	Panel B: Selected Empirical Measures
Spell length	Number of years CEO completes before leaving office
% forced	Percentage of CEO spells that end in CEO dismissal
% forced per year	Percentage of firm-years that end in CEO dismissal
	Panel C: Subsamples
Y1 to Y2	Contains all CEO spells that end between years Y1 and Y2
Small (large) firms	Firm's assets $\leqslant$ (>) \$6.6B, in CPI-adjusted 2007 dollars
Low (high) ownership	Percentage of shares owned by non-CEO officers and directors $\leqslant$ (>) 1.31%
Less (more) outsiders	Percentage of directors who are not also officers $\leqslant$ (>) 72.7%

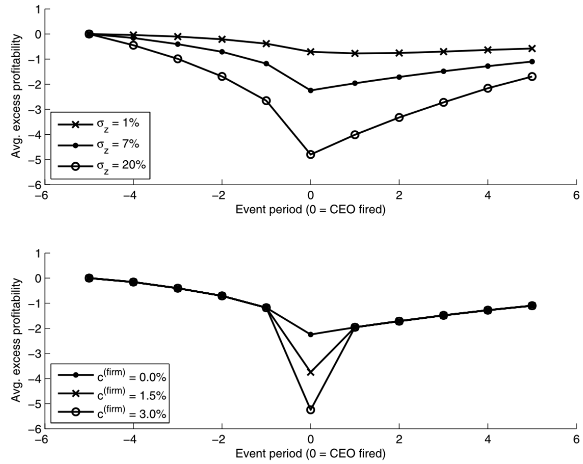

Figure 3. Profitability around CEO dismissals: Comparative statics. This figure shows average excess profitability in event time around CEO dismissals. Variable definitions are in Table I. Results are from 100,000 consecutive CEO spells in a simulated firm. Unless otherwise noted in the legend, simulations use parameter values $\beta = 0.9$, $\mu_0 = 1\%$, $\sigma_0 = 2\%$, $\sigma_\epsilon = 3\%$, $c = 3\%$, $\phi = 0.12$, and $\sigma_z = 7\%$; voluntary turnover occurs after (and only after) completing 15 periods in office. In the top panel there is no firm turnover cost, so $c^{(firm)} = 0$ and $c = c^{(pers)}/\kappa$. In the bottom panel, I hold constant total cost c and vary the firm cost $c^{(firm)}$. To facilitate comparisons, I shift all lines vertically so that they match perfectly at period −5 and value 0%.

- $c^{(\text{firm})}$ and $c^{(\text{pers})}/\kappa$ are the two turnover-cost components; total cost is $c = c^{(\text{firm})} + c^{(\text{pers})}/\kappa$.
- ϕ controls persistence; σ_ε is profitability noise; σ_z governs the precision of the board's extra signal.

5.5 Table II: summary statistics (what the model must match)

Table II provides the empirical targets:

- **Forced turnover is rare:** 2.29% per year in the full sample.
- **Heterogeneity:** forced turnover varies across industries (e.g., chemicals lower; business equipment higher).
- **Time trend:** forced turnover increases in later decades.
- **Spell length:** median spell is longer for unforced exits (median 7 years) than for forced exits (median 4 years).
- **Profitability dispersion:** excess profitability has large cross-sectional variation; this is what the model rationalizes as a mixture of ability and shocks.

5.6 Table III: baseline structural estimates (main quantitative conclusions)

Table III reports the SMM estimates (standard errors in parentheses). Key takeaways:

1. **Firm turnover cost:** $\widehat{c}^{(\text{firm})} = 1.33$ (% of assets). This is the cost that reduces profitability mechanically in the turnover year via (1).

Table II

Summary Statistics

Panel A contains summary statistics on sample size and CEO turnover rates in various subsamples. The full sample consists of CEO spells for firms in the 1970 to 2007 *Forbes* annual compensation surveys. I include complete CEO spells that ended between 1971 and 2006. Additional details are in Section II.A and Table I. The 12 industries are defined on Kenneth French's website. Panel B contains additional statistics for the full sample. Assets is Compustat item AT. ROA is return on assets, computed as Compustat item OIBDP divided by the midpoint of assets. Statistics for spell length weight each CEO equally; other statistics weight each firm-year equally.

Panel A: Sample Size and CEO Turnover Rates

	Firm-Years	CEOs	Forced Successions	Unforced Successions	% Forced	% Forced per Year
Full sample	7,325	981	168	813	17.1	2.29
Consumer nondurables	757	87	13	74	14.9	1.72
Consumer durables	325	42	8	34	19.0	2.46
Manufacturing	1,543	204	28	176	13.7	1.81
Energy	315	40	7	33	17.5	2.22
Chemicals	436	63	3	60	4.8	0.69
Business equipment	605	90	23	67	25.6	3.80
Telecom	153	23	5	18	21.7	3.27
Utilities	662	82	7	75	8.5	1.06
Wholesale and retail	517	88	19	69	21.6	3.68
Health	481	57	8	49	14.0	1.66
Finance	920	127	27	100	21.3	2.93
Other	611	78	20	58	25.6	3.27
1970–1974	580	102	8	94	7.8	1.38
1975–1979	886	132	15	117	11.4	1.69
1980–1984	1,182	146	21	125	14.4	1.78
1985–1989	1,340	163	23	140	14.1	1.72
1990–1994	1,113	126	23	103	18.3	2.07
1995–1999	713	90	20	70	22.2	2.81
2000–2004	989	149	40	109	26.8	4.04
2005–2006	522	73	18	55	24.7	3.45

Panel B: Additional Statistics

Variable	Observations	Mean	Std. dev.	Median	Min	Max
ROA (% p.a.)	7,325	16.0	9.07	15.5	−23.8	85.6
Excess profitability (% p.a.)	7,325	2.00	7.37	0.75	−35.7	68.0
Assets ($billion)	7,325	12.5	55.0	2.38	0.015	1264
Spell length (years):						
All	981	7.5	4.9	6	1	29
Forced	168	5.1	3.8	4	1	21
Unforced	813	8.0	4.9	7	1	29

Table III

Parameter Estimates

This table contains estimates of the parameters from the model in Section I. Estimation uses data on a sample of 981 CEOs, described in Section II.A. Parameters are estimated using SMM, as described in Section II. Parameter definitions are in Table I. Standard errors are in parentheses.

Firm Cost $c^{(firm)}$	Personal Cost $c^{(pers)}/\kappa$	Prior Mean μ_0	Prior Stdev. σ_0	Persistence ϕ	Profit. Stdev. σ_ϵ	z Signal Stdev. σ_z
1.33	4.61	0.88	2.42	0.125	3.43	5.15
(0.61)	(0.58)	(0.34)	(0.06)	(0.004)	(0.09)	(0.33)

Table IV

Moments Used in SMM Estimation

Panel A shows the 14 moments used in the SMM estimation described in Section II. Variable definitions are in Table I. Empirical moments are computed from the sample of 981 CEOs described in Section II.A. Simulated moments are computed from data simulated from the model using parameter values in Table III. Moments' standard errors and p-values are computed by Monte Carlo, as follows. I create 10,000 sets of 14 moments, each from a simulation of 981 CEOs (to match the empirical sample size) using parameter values in Table III. The standard error is the standard deviation of the 10,000 simulated moments, and the p-value is the fraction of simulated moments that are at least as far from the mean simulated moment as is the empirical moment. The first eight moments come from the pooled regression $y^*_{it} = \lambda_0 + \lambda_1 y^*_{i,t-1} + \Delta^{(-2)} + \Delta^{(-1)} + \Delta^{(0)} + \Delta^{(1)} + \Delta^{(2)} + \delta_{it}$. $\Delta^{(k)}$ is a fixed effect for whether forced CEO turnover occurred at the end of period $t + k$. $E[\text{Var}(X)]$ is the mean across CEOs of the within-CEO variance of X_{it}; vice versa for $\text{Var}(E[X])$. Hazard rates $h^{(j)}$ equal the percentage of CEOs forced out of office per year during tenure period (j), conditional on the CEO reaching (j). Panel B shows the χ^2 statistic and corresponding p-value for SMM's test of overidentifying restrictions, which jointly tests whether the empirical and simulated moments are equal, and which is defined in the Internet Appendix.

Panel A: 14 Moments from SMM Estimation

Moment	Notation	Empirical Value	Simulated Value	Standard Error	p-Value
Profitability intercept	λ_0	0.22	0.20	0.04	0.66
Profitability AR(1) coefficient	λ_1	0.87	0.88	0.01	0.38
Δ profitability 2 yrs before	$\Delta^{(-2)}$	−0.37	−0.62	0.27	0.29
Δ profitability 1 yr before	$\Delta^{(-1)}$	−1.36	−1.06	0.27	0.14
Δ profitability 0 yrs before	$\Delta^{(0)}$	−1.73	−1.99	0.26	0.20
Δ profitability 1 yr after	$\Delta^{(1)}$	−1.15	−0.02	0.27	0.00
Δ profitability 2 yrs after	$\Delta^{(2)}$	0.60	0.42	0.27	0.38
Var(Profitability residuals)	$Var(\delta)$	11.99	11.92	0.20	0.60
Forced turn. hazard rate, yrs 1–2	$h^{(1-2)}$	2.52	1.76	0.29	0.00
Forced turn. hazard rate, yrs 3–4	$h^{(3-4)}$	2.66	3.46	0.46	0.03
Forced turn. hazard rate, yrs 5–7	$h^{(5-7)}$	2.10	2.71	0.39	0.10
Forced turn. hazard rate, yrs 8+	$h^{(8+)}$	1.96	1.47	0.26	0.04
Cross-CEO avg. of within-CEO profitability variance	$E[\text{Var}(X)]$	799.3	782.1	73.5	0.77
Cross-CEO variance of within-CEO avg. profitability	$\text{Var}[E(X)]$	221.5	183.5	20.9	0.11

Panel B: Test of Overidentifying Restrictions

$\chi^2 = 33.2$ p-Value= 0.000

2. **Effective personal cost (entrenchment):** $\widehat{c^{(\text{pers})}/\kappa} = 4.61$ (% of assets). This is large relative to $c^{(\text{firm})}$, implying that *board private costs are the dominant reason firing is rare in the baseline fit.*

3. **Total turnover cost:** $\widehat{c} = 1.33 + 4.61 = 5.94$ (% of assets).

4. **Learning/prior:** $\widehat{\mu}_0 = 0.88$, $\widehat{\sigma}_0 = 2.42$. So the CEO pool has meaningful dispersion in ability, which creates an option value to replacement.

5. **Persistence:** $\widehat{\phi} = 0.125$ implies strong persistence (slow mean reversion), which slows learning from profitability and produces gradual post-firing recoveries.

6. **Noise and extra signal:** $\widehat{\sigma}_\varepsilon = 3.43$ and $\widehat{\sigma}_z = 5.15$. The extra signal is fairly noisy, so profitability remains an important learning channel.

5.7 Table IV: the 14 moments and model fit (what is matched well vs poorly)

Panel A lists each moment, the empirical value, the simulated value under $\widehat{\theta}$, its Monte Carlo standard error, and a Monte Carlo p-value.

How to read the moments:

- λ_0, λ_1 (profitability intercept and AR(1)): simulated values are close to empirical, indicating the model matches average profitability and persistence reasonably well.
- Event-time coefficients $\gamma_{-2}, \gamma_{-1}, \gamma_0, \gamma_1, \gamma_2$: these encode *how profitability moves before and after a firing*. The paper highlights that some of these are hard to fit perfectly. Notably, γ_1 (1 year after firing) differs sharply between data and model (p-value essentially 0), indicating a mismatch in immediate post-dismissal profitability dynamics.
- $\text{Var}(\delta)$ and $E[\text{Var}(X)]$ are matched well, suggesting the model captures *within-CEO* profitability volatility.

- The hazard-rate moments are informative about turnover costs. Some hazard-rate moments show statistically meaningful gaps.

Panel B reports the overidentifying-restrictions test ($\chi^2 = 33.2$ with p-value essentially 0), so the model is *statistically rejected* when all 14 moments are tested jointly. Interpreting this correctly:

- Rejection is common in tight structural models with many moments; the question becomes *which dimensions are matched economically well*, and what the deviations suggest about missing mechanisms.
- Here, the main tension is in some event-time and hazard-shape moments (the model is parsimonious and imposes structure on learning and costs).

5.8 Table V and Figure 4: additional fit and the core empirical patterns

Table V. The model matches key unconditional objects:

- median forced tenure (4 years) and unforced tenure (7 years) exactly,
- % forced and forced per-year rates closely (16.2% vs 17.1%; 2.16% vs 2.29%),
- the probit slope of forced turnover on profitability is negative in both data and model (model slightly smaller in magnitude).

The pseudo-R^2 is low in both data and model, consistent with turnover being driven by noisy signals and latent information.

Table V

Additional Statistics on Model Fit

This table compares statistics computed from the empirical sample (containing 981 CEOs, described in Section II.A) and a sample of 100,000 CEOs simulated from the model using parameter estimates in Table III. Variable definitions are in Table I. In the probit model, the dependent variable is an indicator for whether year t ended in a CEO dismissal, and the independent variable is firm-specific profitability y_t^*. "Slope" is the estimated slope on y_t^*, and "Stderr." is its robust standard error.

	Median Spell Length				Probit Model		
	Unforced	Forced	% Forced	% Forced per Year	Slope	Stderr.	Pseudo-R^2
Empirical	7	4	17.1	2.29	−0.168	[0.026]	0.03
Simulated	7	4	16.2	2.16	−0.125	[0.002]	0.02

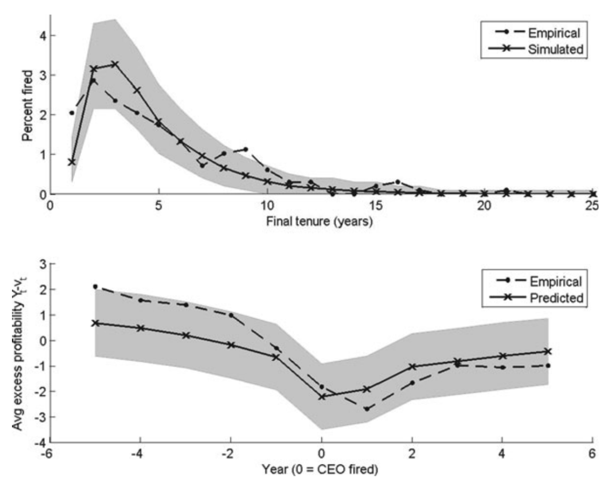

Figure 4. Empirical and predicted patterns in CEO dismissals and profitability. The top panel shows the unconditional percentage of CEOs fired at different tenure levels. The bottom panel shows average excess profitability in event time around dismissals. In both panels, the empirical pattern is computed from the sample of 981 CEO spells described in Section II.A. The predicted pattern is the average across 10,000 simulated samples, each of which contains 981 CEOs. Simulations use parameter values from Table III. The grey 95% confidence region covers the area between the 2.5% and 97.5% percentiles from the 10,000 simulated samples.

Figure 4. Figure 4 compares empirical patterns to model predictions:

- **Top panel:** dismissal probability vs final tenure is broadly hump-shaped and then declines; the model replicates the broad level and shape within uncertainty bands.
- **Bottom panel:** profitability falls leading up to dismissal and then recovers; the predicted path is qualitatively aligned with the empirical path, though not perfect in the immediate post-dismissal period (consistent with Table IV).

5.9 Table VI: subsample estimation (governance interpretation)

Table VI estimates the model separately in subsamples and compares fit.

Panel A (parameters). Key patterns:

- **Small vs large firms:** small firms have very high estimated effective personal cost (8.53), while large firms have an estimate near zero and slightly negative (−0.29). A negative effective personal cost means boards (in that subsample) appear to fire *more* than the shareholder-value-maximizing benchmark would predict ex post, which the paper discusses.
- **Early vs late period:** the effective personal cost falls sharply from early years (8.32) to later years (2.28), consistent with a secular decline in entrenchment.
- **Board structure:** "more outsiders" is associated with a lower effective personal cost (3.00) than "less outsiders" (8.25), consistent with outsider-dominated boards monitoring more aggressively.

Panel B (fit). The model generally preserves the same qualitative mapping between profitability and firing within subsamples (probit slopes negative), but the magnitudes vary.

5.10 Table VII: counterfactuals (how costly is entrenchment?)

Table VII runs two experiments.

Experiment 1: eliminate entrenchment. This sets $c^{(\mathrm{pers})} = 0$ while holding other parameters fixed (within each subsample).

The mechanics:

- firing rates jump dramatically (e.g., full sample from 2.1% to 12.3% fired per year),
- CEO tenures fall (full sample from 7.5 to 4.5 years),
- mean profitability rises only modestly (full sample from 15.5 to 16.0),
- shareholder value rises meaningfully (full sample +3.1%; larger gains in some subsamples).

The key economic insight is that *even large changes in firing frequency translate into relatively small changes in mean profitability*, because (i) learning takes time and (ii) persistence implies the firm cannot instantly recover from past mistakes.

Experiment 2: hire a very bad CEO. This evaluates how much shareholder value falls when the firm is unlucky and draws an extremely low-ability CEO first. The comparison illustrates that a high effective personal cost can exacerbate losses from a bad CEO by delaying replacement.

Table VI

Estimation in Subsamples

Each pair of rows contains estimation results using a subset of the 981 CEOs from the full sample. Variable and subsample definitions are in Tables I, IV, and V. Panel A contains parameter estimates with standard errors in parentheses, the number of CEO spells used, and the χ^2 and p-value from the test of overidentifying restrictions. Panel B compares statistics computed from the empirical subsample and a sample simulated from the model using parameters from Panel A.

Panel A: Parameter Estimates in Subsamples

	Firm Cost $c^{(firm)}$	Personal Cost $c^{(pers)}/\kappa$	Prior Mean μ_0	Prior Stdev. σ_0	Persis-tence ϕ	Profit. Stdev. σ_ϵ	z Signal Stdev. σ_z	CEOs	χ^2 (p-val.)
Small firms	0.01	8.53	1.38	3.26	0.120	4.01	5.34	543	18.7
	(1.05)	(1.06)	(0.69)	(0.05)	(0.004)	(0.14)	(0.24)		(0.01)
Large firms	1.71	−0.29	0.35	1.23	0.135	2.89	7.65	438	29.9
	(0.64)	(0.74)	(0.37)	(0.03)	(0.002)	(0.10)	(0.27)		(0.00)
1971–1989	0.05	8.32	0.49	2.52	0.125	3.30	4.37	222	21.3
	(0.91)	(1.02)	(0.44)	(0.05)	(0.003)	(0.10)	(0.20)		(0.00)
1990–2006	1.67	2.28	1.24	2.72	0.123	3.61	9.51	222	27.5
	(0.70)	(0.76)	(0.67)	(0.07)	(0.002)	(0.14)	(0.46)		(0.00)
Low ownership	0.04	6.41	−0.76	3.10	0.126	3.10	12.92	327	16.1
	(1.07)	(1.09)	(0.60)	(0.11)	(0.003)	(0.11)	(0.93)		(0.02)
High ownership	0.00	7.99	0.14	3.99	0.088	3.35	8.62	325	8.6
	(1.19)	(1.54)	(0.94)	(0.18)	(0.004)	(0.15)	(1.03)		(0.29)
Less outsiders	0.00	8.25	1.57	2.93	0.117	3.69	5.84	491	9.7
	(1.26)	(1.69)	(0.67)	(0.11)	(0.004)	(0.18)	(0.94)		(0.21)
More outsiders	0.00	3.00	−0.56	1.98	0.116	2.87	10.80	490	14.7
	(0.85)	(0.99)	(0.47)	(0.07)	(0.002)	(0.10)	(0.65)		(0.04)

Panel B: Model Fit in Subsamples

		Median Tenure					Probit Model		
		Unforced	Forced	% Forced	% Forced/yr	$\sigma(E[X])$	Slope	Stderr.	Pseudo-R^2
Small firms	Data	7	4	15.7	2.14	19.1	−0.143	0.029	0.04
	Model	7	3	16.6	2.23	17.1	−0.089	0.002	0.01
Large firms	Data	7	4	18.6	2.45	10.3	−0.207	0.050	0.03
	Model	7	5	19.1	2.59	10.0	−0.252	0.003	0.04
1971–1989	Data	7	5	12.3	1.68	14.6	−0.121	0.031	0.03
	Model	7	4	10.2	1.30	12.7	−0.072	0.002	0.02
1990–2006	Data	7	4	23.1	3.03	16.1	−0.242	0.045	0.04
	Model	7	4	23.2	3.27	15.4	−0.225	0.002	0.04
Low ownership	Data	8	5	12.6	1.51	12.3	−0.161	0.040	0.06
	Model	7	5	12.1	1.54	12.1	−0.161	0.002	0.08
High ownership	Data	8	4	17.6	2.14	24.6	−0.121	0.031	0.03
	Model	7	4	16.6	2.22	19.8	−0.072	0.002	0.02
Less outsiders	Data	7	4.5	11.6	1.58	17.9	−0.080	0.038	0.02
	Model	7	4	12.0	1.54	14.8	−0.074	0.002	0.02
More outsiders	Data	8	5	16.0	2.05	10.4	−0.200	0.043	0.05
	Model	7	5	13.7	1.77	10.7	−0.171	0.002	0.05

Table VII

The Effect of Entrenchment on Shareholder Value

Each row presents results from a different counterfactual experiment in which I change one or more model input. Variable and subsample definitions are in Table I. All results are from simulations of the model described in Section I. "Treat." denotes the treatment sample. Experiment 1: The control uses the denoted subsample's parameter estimates from Tables III or VI, and treatment uses those same parameters but sets $c^{(pers)}$ to zero. Experiment 2: Both the control and treatment use full-sample parameter estimates from Table III with effective personal cost as noted below. In the control, all CEOs' abilities are drawn from the prior distribution. In the treatment for experiment 2, the firm hires a CEO with ability two standard deviations below the mean, so $\alpha = \mu_0 - 2\sigma_0$, and subsequent CEOs' abilities are drawn from the prior distribution. "% effect on shareholder value" is the percentage difference in NPV between treatment and control, averaging across all years (for experiment 1), or measured when the first CEO is hired (experiment 2). Other column values are averages across all simulation years (experiment 1), or across the first CEOs' years (experiment 2).

	% of CEOs Fired per Year		Mean Length (Years)		Mean Profitability (%/Year)		% Effect on Shareholder value
	Control	Treat.	Control	Treat.	Control	Treat.	
Experiment							
1. Eliminate entrenchment							
Full sample	2.1	12.3	7.5	4.5	15.5	16.0	+3.1
1971–1989	1.3	25.7	7.9	2.9	15.2	16.2	+6.9
1990–2006	3.3	8.7	7.1	5.4	15.8	16.0	+1.4
Low ownership	1.6	25.1	7.8	3.0	13.9	14.6	+5.1
High ownership	2.2	26.3	7.5	2.9	15.1	16.4	+8.8
Few outsiders	1.6	26.4	7.8	2.9	16.3	17.4	+6.5
More outsiders	1.8	25.4	7.7	3.0	14.0	14.3	+2.5
Small firms	2.3	26.6	7.4	2.8	16.3	17.6	+7.7
Large firms	2.6	1.8	7.4	7.7	14.6	14.6	+0.0
2. Hire a very bad CEO							
Pers. cost = estimate	2.1	17.3	7.5	4.0	15.5	13.9	−6.3
Pers. cost = zero	12.3	57.1	4.5	1.6	16.0	14.4	−4.6

Table VIII

Alternate Specifications

Each pair of rows contains estimation results (parameter estimates, standard errors in parentheses, and statistics for the test of overidentifying restrictions) from a different specification of the model. Variable definitions are in Table I. "Main results" are the same as in Tables III and IV. "Alt. forced def'n" assumes CEO successions are forced if and only if the CEO is aged 64 or less at the time of succession. "Fixed effects" uses profitability data de-meaned at the firm level. "30 industries" computes industry-adjusted profitability using the 30 industries defined on Kenneth French's website. "Flat threshold" forces the firing threshold to stay constant with tenure. "$c_{(retire)}$ = 0" sets the turnover cost of voluntary succession to zero. The last rows use different values of β.

	Firm Cost $c^{(firm)}$	Personal Cost $c^{(pers)}/\kappa$	Prior Mean μ_0	Prior Stdev. σ_0	Persis-tence ϕ	Profit. Stdev. σ_ϵ	z Signal Stdev. σ_z	χ^2 (p-val.)
Main results	1.33	4.61	0.88	2.42	0.125	3.43	5.15	33.2
	(0.61)	(0.58)	(0.34)	(0.06)	(0.004)	(0.09)	(0.33)	(0.00)
Alt. fired def'n	0.00	1.34	1.46	1.24	0.125	3.45	6.08	398.6
	(0.41)	(0.34)	(0.38)	(0.03)	(0.003)	(0.08)	(0.33)	(0.00)
Fixed effects	1.00	5.13	−0.10	1.61	0.262	3.31	2.90	39.1
	(0.53)	(0.55)	(0.19)	(0.03)	(0.004)	(0.08)	(0.15)	(0.00)
30 industries	0.86	4.05	0.89	2.30	0.134	3.46	6.29	38.7
	(0.59)	(0.52)	(0.40)	(0.05)	(0.002)	(0.09)	(0.19)	(0.00)
Flat threshold	0.71	5.28	0.87	2.94	0.132	3.32	6.74	33.1
	(0.67)	(0.52)	(0.34)	(0.05)	(0.004)	(0.11)	(0.47)	(0.00)
$c_{(retire)} = 0$	0.01	5.94	0.67	3.34	0.128	3.42	9.71	36.0
	(0.57)	(0.53)	(0.32)	(0.04)	(0.001)	(0.08)	(0.35)	(0.00)
$\beta = 0.85$	1.21	2.68	0.81	2.47	0.129	3.45	6.30	37.2
	(0.57)	(0.60)	(0.38)	(0.04)	(0.002)	(0.08)	(0.23)	(0.00)
$\beta = 0.95$	1.60	5.33	1.00	2.01	0.129	3.37	5.06	43.3
	(0.52)	(0.56)	(0.39)	(0.05)	(0.003)	(0.11)	(0.37)	(0.00)

5.11 Table VIII: robustness

Table VIII re-estimates under alternative assumptions:

- Alternative forced-turnover definition changes the implied entrenchment cost substantially.
- Adding firm fixed effects or changing the industry benchmark alters persistence and signal precision estimates (e.g., ϕ rises in the fixed-effects specification).
- Forcing a flat (tenure-invariant) firing threshold changes some estimates but does not remove the need for sizable turnover costs.
- Setting voluntary turnover cost to zero (costless retirement) shifts parameter magnitudes but preserves the basic interpretation that turnover is costly.
- Varying β changes the implied total turnover cost and its decomposition; but turnover costs remain economically meaningful.

6 Synthesis: the answer to "why are CEOs rarely fired?"

Putting the model and estimates together:

1. **Firing is expensive.** Even ignoring entrenchment, the firm faces a real turnover cost.
2. **Boards face additional private costs.** The estimated effective personal cost is large in the baseline fit, which rationalizes low firing frequencies and creates CEO entrenchment.

3. **Learning is slow because performance is persistent and noisy.** With ϕ small and σ_ε large, one year of bad performance is not enough to precisely infer low ability.

4. **Therefore,** boards wait for multiple negative signals before paying turnover costs, so dismissals cluster after sustained underperformance and are rare on an annual basis.

7 Conclusion

7.1 Summary

This paper answers a concrete benchmark question: *is the observed forced CEO turnover rate (about 2% per year) "low" in an economically meaningful sense?* It concludes **yes**, in the sense that matching the observed firing frequency in a disciplined dynamic model requires **very large effective turnover costs**. Moreover, the paper concludes that **most of this effective cost is not a real disruption cost borne by shareholders, but rather an entrenchment wedge** that makes boards behave as if firing is much more expensive than it truly is.

The argument is one chain:

1. **Dynamic benchmark:** a rational board chooses each year whether to keep or replace the CEO while **learning about CEO ability** from (i) profits and (ii) an additional signal. Replacement is costly, so firing occurs only when the *expected gain* from a new CEO (net of learning) exceeds the *turnover cost.*

2. **Structural estimation:** the model is estimated by SMM on CEO spell data and profitability moments, so parameter magnitudes are disciplined by (i) the low observed firing rate, (ii) profitability dynamics, and (iii) event-time patterns around dismissals.

3. **Main quantitative result (turnover costs):** the estimated *real* turnover cost is

$$\widehat{c}^{(\text{firm})} \approx 1.33\% \text{ of assets,}$$

   but the estimated *effective board personal turnover cost* (identified as a ratio) is

$$\widehat{c^{(\text{pers})}/\kappa} \approx 4.61\% \text{ of assets.}$$

   Hence, the board behaves *as if* total shareholder turnover cost were

$$\widehat{c}^{(\text{firm})} + \widehat{c^{(\text{pers})}/\kappa} \approx 5.9\% \text{ of assets,}$$

   even though shareholders actually pay only the firm component $\widehat{c}^{(\text{firm})}$. In dollar terms (using the paper's scaling), the effective personal cost is roughly **$200 million for the median firm**. This gap is interpreted as **CEO entrenchment**: boards retain some CEOs whom shareholders would prefer to replace ex post.

4. **Value implications:** a counterfactual experiment sets the entrenchment wedge to zero ($c^{(\text{pers})} = 0$), holding everything else fixed. The model predicts:

   - forced turnover rises from about **2.1%** to **12.3%** per year,
   - mean CEO tenure falls from about **7.5** to **4.5** years,
   - mean profitability rises modestly (about **+0.5** percentage points per year),
   - and shareholder value (proxied by discounted future profits / market-to-book) rises by about **3.1%**.

   The modest profitability gain despite a large rise in firing frequency is a key conclusion: even without entrenchment, (i) learning takes time, (ii) turnover triggers real disruption costs more often, and (iii) the very worst CEOs were already being fired.

5. **Governance interpretation (supporting evidence):** estimated entrenchment is systematically higher in subsamples commonly viewed as having weaker governance (e.g., earlier decades, fewer outside directors, smaller firms), providing an internal consistency check for the interpretation of the personal turnover cost.

7.2 Why CEOs are rarely fired in the model (and in the data)

The paper's intuition is that **rare firing is the equilibrium outcome of a learning problem with (perceived) large adjustment costs**.

1) Learning makes "one bad year" insufficient. Boards do not directly observe CEO ability α; they infer it from noisy and persistent performance. When profitability is persistent (low mean reversion) and noisy, it takes multiple periods to be confident that poor performance reflects low ability rather than bad luck. So even an untalented CEO can survive several years before the posterior belief falls below the firing threshold.

2) Turnover costs rationally create an inaction region. Replacing a CEO has an immediate cost. Thus there is an *inaction band*: as long as the expected value of keeping the incumbent is close to the value of switching (net of cost), the board rationally waits.

3) Entrenchment shifts the firing threshold downward. With a board personal cost, the board behaves as if firing costs shareholders

$$c^{(\text{firm})} + c^{(\text{pers})}/\kappa,$$

even though shareholders bear only $c^{(\text{firm})}$. This *wedges* the board's decision rule away from shareholder-optimal replacement, making firings rarer and delayed, i.e., **ex post entrenchment**.

4) Why the welfare gain from eliminating entrenchment is "only" a few percent. Even if entrenchment disappears, the firm still must (i) learn, and (ii) pay real disruption costs more often. These forces limit the net profitability improvement and therefore limit the value gain, despite a large increase in turnover.

7.3 Key caveat

The paper's welfare counterfactual is *partial equilibrium* in the sense that it holds other primitives fixed. The paper emphasizes that entrenchment can be **bad ex post** but not necessarily **bad ex ante**: shareholders might rationally choose governance structures that tolerate some CEO protection to attract or retain higher-quality CEOs. A full normative conclusion would require endogenizing governance choice and the CEO talent pool.

7.4 Takeaway

Taylor (2010) shows that explaining the observed ∼2% annual CEO firing rate in a dynamic learning model requires boards to behave as if turnover costs are extremely large (about 5.9% of assets), and that most of this is an entrenchment wedge rather than a real shareholder cost; removing this wedge would sharply raise firing frequency but increase shareholder value by only about 3% because learning and real turnover costs limit the net gains.